

TECHNICAL REPORT

Improving Person-Level External Transfer on V-COCO

A locked study of person scale, view conditioning, factorized labels, representation choice, and selective reliability

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<https://github.com/abdullahuseyinli-dot/human-activity-classification>

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Abstract

The original POLAR study reached 0.961 three-class macro-F1 in-domain but transferred to V-COCO at 0.667 image-level and 0.684 person-level macro-F1. This follow-up isolates the source of that gap under a split-preserving V-COCO protocol. The study controls the person crop, context width, aspect-ratio handling, representation, classifier, factorized posture and motion targets, partial fine-tuning, augmentation, background interventions, and person-box geometry. All choices were made on the official training and validation splits. The selected system and evaluation code were hash-locked before the official test labels were opened once.

The selected scale-conditioned stack combines DINOv2-B probabilities from a tight person view and a 25% context view with five geometry features. It reached 0.8594 validation macro-F1 and 0.8663 official-test macro-F1. On the same 6,077 test people, the historical source-only DINO system reached 0.7071. The paired improvement was +0.1592 macro-F1 with a 95% image-cluster bootstrap interval of [+0.1454, +0.1735]. Accuracy improved from 0.7010 to 0.8795, log loss fell from 1.6525 to 0.2902, and expected calibration error fell from 0.2288 to 0.0076. The largest test gains occurred for short and small-box people. A factorized posture-motion classifier improved over a flat classifier using exactly the same inputs by +0.0111 validation macro-F1, while AugMix and top-block fine-tuning were statistically tied with the frozen multiview stack. These results identify person scale, view construction, and label structure as the main recoverable components of the original transfer gap.

1. Research questions

The experiments answer five questions:

- 1 How much of the external gap comes from showing the model the wrong spatial view?
- 2 Does a modern frozen representation transfer better than a fully adapted source classifier?
- 3 Does the non-exclusive structure of V-COCO actions benefit from a posture-motion factorization?
- 4 Do partial fine-tuning and stronger augmentation add reliable value beyond the frozen representation?
- 5 Do improvements hold on the untouched official test split, including small people, crowded images, and person boxes that touch an image boundary?

The endpoint is three-class person-level classification: sitting, standing, and walking/running. V-COCO's original endpoint is visual semantic role labeling, so its role average precision is a different measurement. Comparisons in this report use one fixed person-level mapping and one locked split protocol.

2. Data and protocol

2.1 Split construction

V-COCO provides person-centric action annotations over COCO images. The study keeps the official train, validation, and test memberships. A deterministic action-tag mapping creates the three target classes while preserving posture, motion, gait, and source co-tags in separate columns. People with incompatible sitting and standing tags are kept visible in the audit and excluded from the mutually exclusive endpoint.

Split	People	Source images	Sitting	Standing	Walking/running
Train	3,090	1,924	923	1,576	591
Validation	3,550	2,199	1,049	1,831	670
Official test	6,077	3,708	1,882	2,962	1,233

The cross-dataset audit found no confirmed source-related pair between the cleaned POLAR data and the retained V-COCO development or test images. Sixty test images that overlapped the legacy external evaluation were quarantined before model fitting. The resulting protocol lock has SHA-256 digest 3a90d6720a6cf5250b995820801199eca611706d514e7b5de2c83bea03f5a143.

2.2 Selection and test access

Models were fitted on the official training split and selected on validation. After selection, the fixed stack was refitted on all 6,640 development people in 4,123 images. Base learners were cross-fitted by source image so that the stacker saw out-of-fold base predictions. The selected artifact, inference dependencies, metric code, historical checkpoints, and development evidence were bound into selection lock 4c0d13c05d537e28066000092e88c37f66b937111516715c8f1756cab162e10d.

The test feature pass read image and box metadata without reading a label column. The evaluation stage then opened the official test labels once, evaluated the selected stack and the historical baseline, and wrote a persistent access gate. No test result was used for model choice, thresholding, calibration, class weighting, or blending.

2.3 Metrics and uncertainty

Macro-F1 is the primary metric. Secondary metrics are accuracy, balanced accuracy, weighted F1, multiclass log loss, Brier score, fixed-bin ECE, adaptive ECE, classwise ECE, class-level precision/recall/F1, confusion matrices, and risk-coverage area. The paired confidence interval resamples 3,708 source images and retains every annotated person belonging to each sampled image. The confirmatory rule required at least +0.01 macro-F1 and a 95% paired interval above zero.

3. Methods

3.1 Historical source-only baseline

The baseline is the three-seed mean of the original POLAR DINOv2-B classifier with its top four transformer blocks adapted. Its four-class probabilities are collapsed by summing walking and running. The original 25% context crop and center-crop preprocessing are replayed exactly; validation probabilities match the earlier artifact to numerical precision.

3.2 Person views and preprocessing

Five deterministic views were studied: full frame, tight person box, and person boxes with 10%, 25%, or 50% context. Aspect-preserving evaluation pads the crop to a square before bicubic resizing to 224 pixels. This retains the whole person and avoids the limb truncation introduced by resizing followed by a center crop.

Two background interventions keep the person pixels unchanged while either blurring or masking the background inside the 25% context crop. They distinguish useful person scale from useful background content.

3.3 Representations and classifiers

The controlled screen uses revision-pinned frozen features from DINOv2-B, ConvNeXt-S, and SigLIP2-B. Logistic regression and calibrated linear SVMs are compared with and without class weighting and with visual-only or visual-plus-geometry inputs. The five geometry variables are log box area, log aspect ratio, normalized box center x and y, and log person height. Every cross-validation fold groups people by source image.

The selected base learners are unweighted multinomial logistic regressions with $C=0.01$, fitted separately to tight and 25%-context DINOv2-B features. Their cross-fitted log probabilities and the five geometry variables feed an unweighted multinomial stacker with $C=1.0$.

3.4 Factorized posture and motion

The flat three-class target is decomposed into seated versus upright posture and, conditional on upright posture, stationary versus locomoting motion. The decoder reconstructs sitting, standing, and walking/running probabilities. A matched flat classifier uses the same tight feature, context feature, and geometry inputs, making the factorization comparison controlled.

3.5 Partial fine-tuning and augmentation

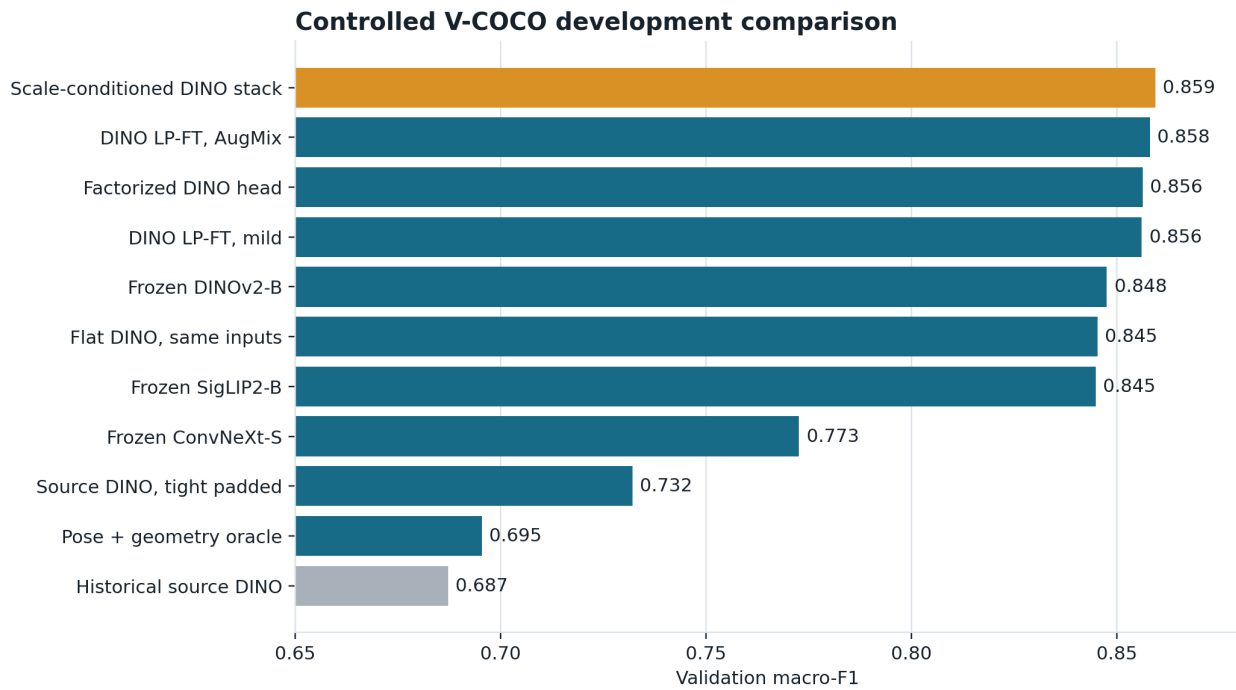
Neural adaptation follows a linear-probe-then-fine-tune schedule. A frozen DINOv2-B probe is trained first, its exact checkpoint initializes the next stage, and only the top transformer block and classifier are updated. Dropout is 0.10. The person-safe mild policy uses horizontal flips, small affine changes, restrained color jitter, low-amplitude Gaussian noise, and small random erasing after square padding. The AugMix variant adds severity-2 AugMix while retaining the same bounded geometry and erasing policy. Training stops from validation performance under the locked epoch budget.

3.6 Diagnostic controls

Ground-truth COCO keypoints plus geometry feed a calibrated RBF SVM as a pose oracle. Because the keypoints come from the target annotation source, this is a diagnostic upper-information control rather than a deployment candidate. Few-shot curves sample complete source-image groups for per-class label budgets of 1, 5, 10, 25, 50, and 100, with 20 deterministic repeats per budget.

4. Development results

Candidate	Validation macro-F1	Accuracy	Log loss	Role
Scale-conditioned DINO stack	0.8594	0.8777	0.2983	Selected
DINO LP-FT, AugMix	0.8580	0.8752	0.3251	Neural adaptation
DINO LP-FT, mild	0.8561	0.8718	0.3284	Neural adaptation
Factorized DINO head	0.8563	0.8758	0.3096	Structured target
Frozen DINO, tight view	0.8476	0.8696	0.3149	Best single view
Frozen SigLIP2	0.8449	0.8662	0.3311	Representation control
Frozen ConvNeXt	0.7726	0.7834	0.5153	Representation control
Source DINO, tight padded view	0.7321	0.7285	1.4882	Source-only weights
COCO pose and geometry oracle	0.6954	0.7166	0.6278	Diagnostic
Historical source DINO	0.6872	0.6780	1.8448	Baseline



Controlled V-COCO development comparison

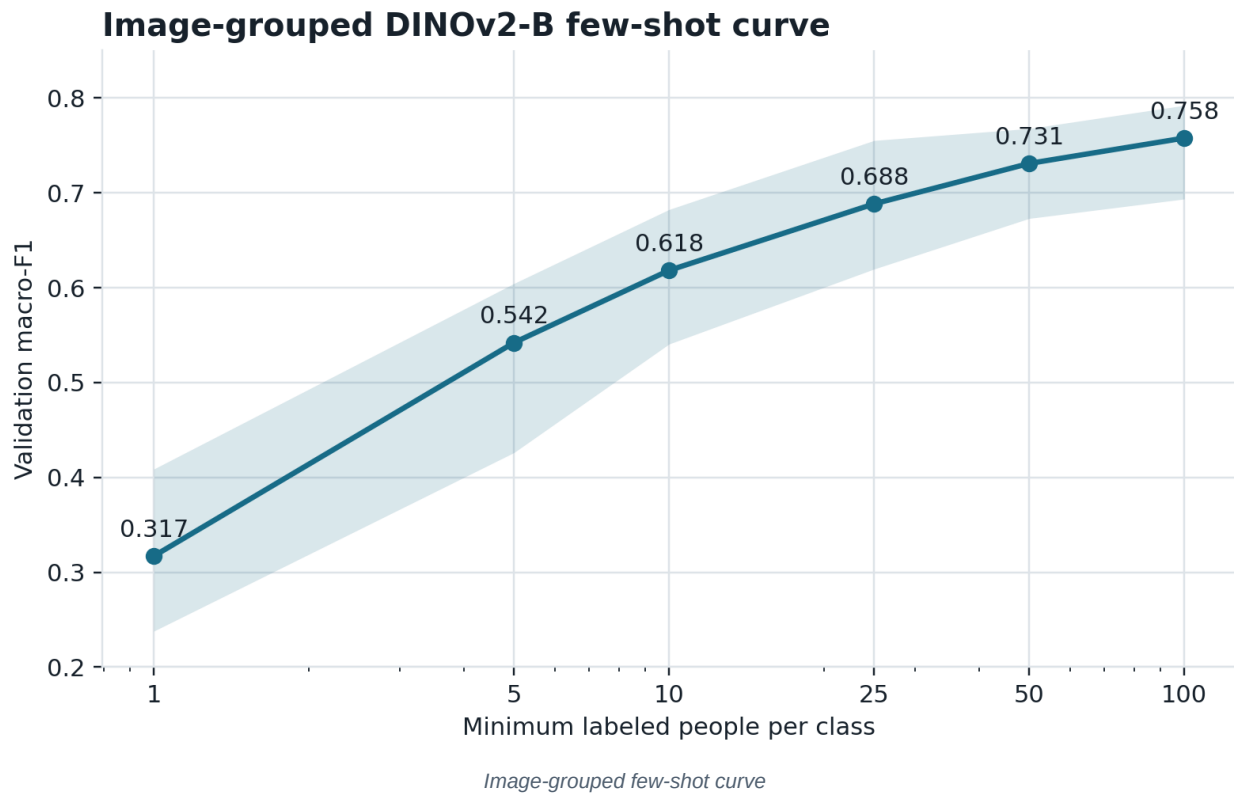
The stack improved over the historical validation baseline by +0.1722 macro-F1, with a 95% image-cluster interval of [+0.1538, +0.1907]. It also improved over the best single-view DINO control by +0.0118, interval [+0.0037, +0.0201], satisfying the predeclared promotion rule.

The factorized head reached 0.8563 versus 0.8453 for the flat classifier with the same inputs. The paired difference was +0.0111, interval [+0.0056, +0.0166]. This isolates a benefit from target structure rather than backbone, crop, or feature differences.

AugMix improved over mild partial fine-tuning by +0.0020, but its interval [-0.0036, +0.0075] included zero. The stack and AugMix model were likewise tied: +0.0014, interval [-0.0075, +0.0102]. Additional neural seeds were not promoted because the observed margin was below the +0.01 rule.

4.1 Few-shot behavior

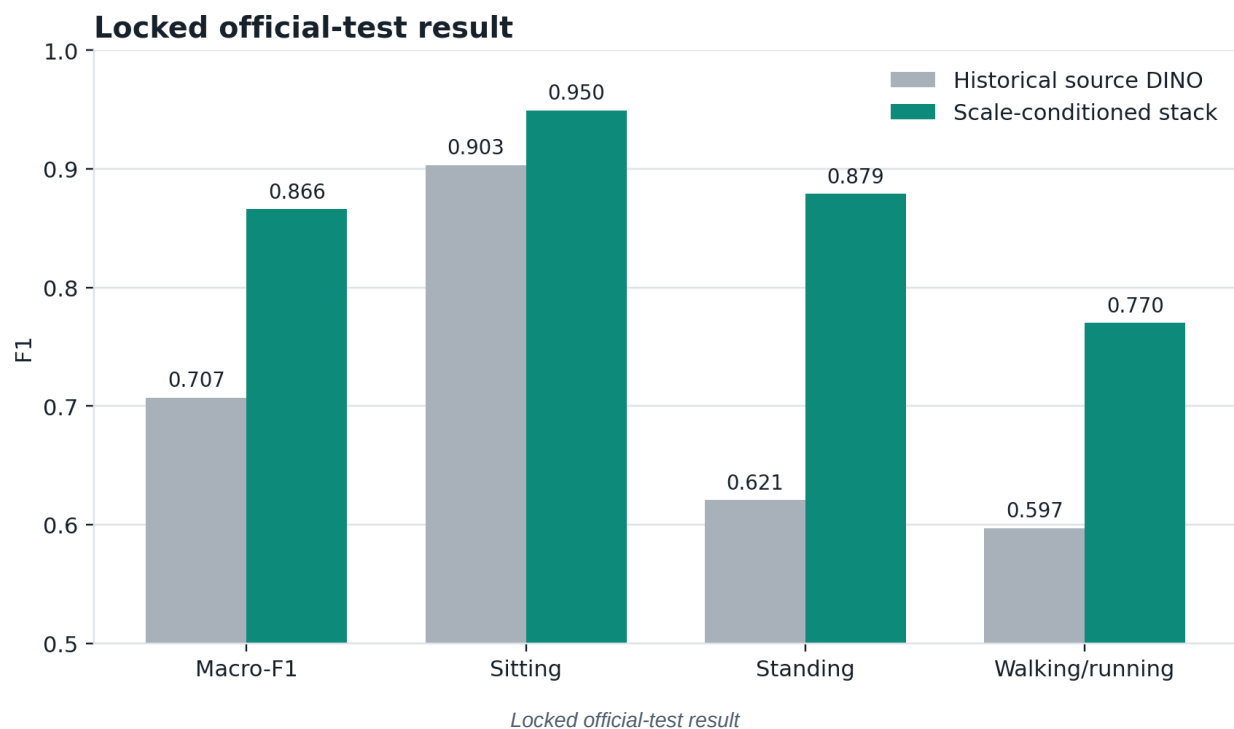
People per class	Mean macro-F1	Standard deviation
1	0.3170	0.0768
5	0.5419	0.0555
10	0.6181	0.0407
25	0.6882	0.0434
50	0.7308	0.0298
100	0.7575	0.0275



The first 25 labels per class recover the historical source-only validation result on average. Performance continues to rise through 100 labels per class, so the remaining gap is not exhausted by the measured few-shot range.

5. Official test results

Method	Macro-F1	Accuracy	Balanced accuracy	Log loss	ECE	AURC
Scale-conditioned DINO stack	0.8663	0.8795	0.8636	0.2902	0.0076	0.0296
Historical source DINO	0.7071	0.7010	0.7674	1.6525	0.2288	0.2007



The paired macro-F1 gain is +0.1592, with a 95% interval of [+0.1454, +0.1735]. The confirmatory rule passed. Validation and test macro-F1 differ by only +0.0069, while test log loss is slightly lower, showing that the locked choice generalized without a drop at the final gate.

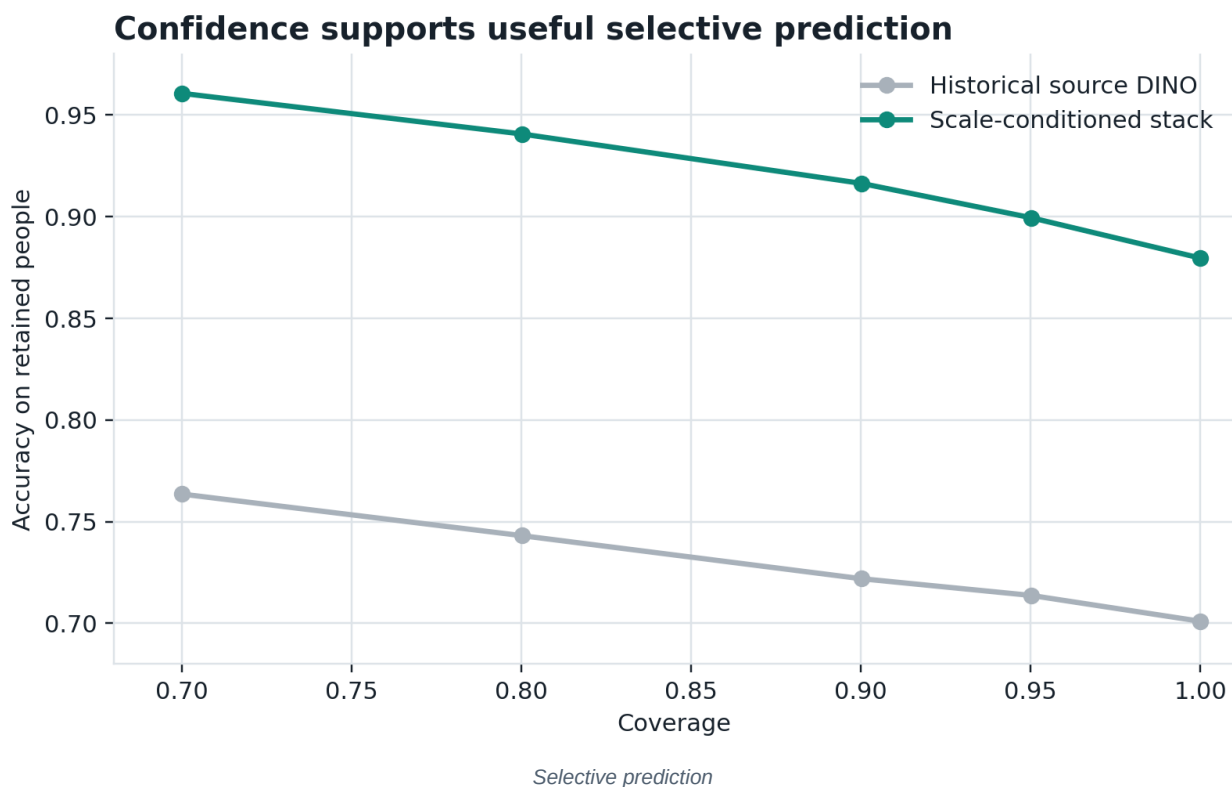
5.1 Per-class behavior

Class	Stack precision	Stack recall	Stack F1	Baseline F1	F1 change
Sitting	0.9438	0.9554	0.9496	0.9033	+0.0462
Standing	0.8731	0.8852	0.8791	0.6209	+0.2582
Walking/running	0.7913	0.7502	0.7702	0.5970	+0.1732

The historical classifier predicts walking/running for 1,355 standing people. The stack reduces that error to 236. It makes the opposite error for 305 locomoting people, up from 98 in the baseline, so the remaining boundary is asymmetric: locomotion recall is the principal class-level opportunity.

5.2 Calibration and selective prediction

The stack's ECE is 0.0076, adaptive ECE is 0.0104, and classwise ECE is 0.0098. At 90% coverage it retains 5,470 people, reaches 0.9163 accuracy and 0.9011 macro-F1, and has 8.37% risk. At 70% coverage, accuracy reaches 0.9605 and macro-F1 0.9459. The full risk-coverage area falls from 0.2007 for the baseline to 0.0296 for the stack.



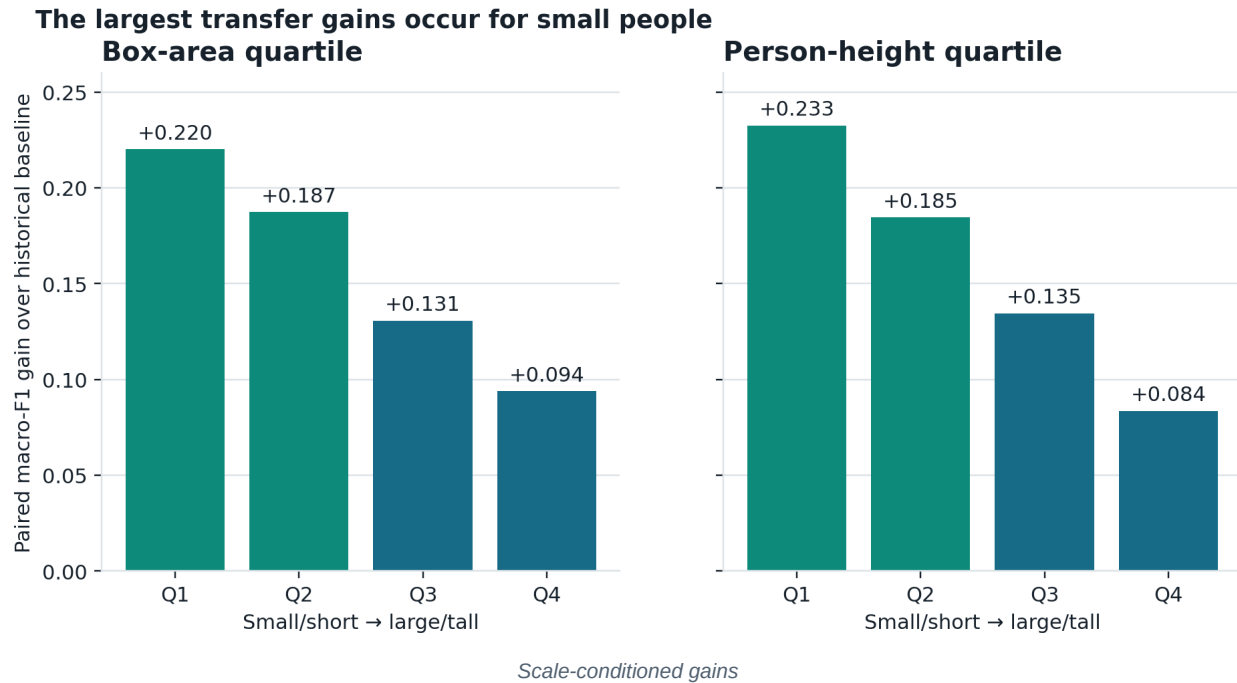
6. What explains the gain

6.1 Person scale is the strongest measured mechanism

The improvement is largest where the historical system is weakest.

Test stratum	Stack macro-F1	Baseline macro-F1	Gain
Smallest box-area quartile	0.8870	0.6668	+0.2202
Area quartile 2	0.8656	0.6782	+0.1874

Test stratum	Stack macro-F1	Baseline macro-F1	Gain
Area quartile 3	0.8485	0.7177	+0.1308
Largest box-area quartile	0.8535	0.7597	+0.0938
Shortest-person quartile	0.8820	0.6495	+0.2325
Tallest-person quartile	0.8291	0.7455	+0.0836



On validation, the correlation between correctness and person height falls from Spearman 0.1503 for the historical model to 0.0061 for the stack; the corresponding box-area correlation falls from 0.1679 to 0.0320. The scale-conditioned system is not merely better on average: it removes most of the historical dependence on how large a person appears.

6.2 Crop construction matters before weight adaptation

Applying a tight, aspect-preserving person view to unchanged source weights raises validation macro-F1 from 0.6872 to 0.7321, a paired gain of +0.0449 with interval [+0.0343, +0.0557]. Frozen target-trained DINO features then reach 0.8476. The background-mask and background-blur controls remain around 0.843, offering no improvement over the ordinary context view. The useful intervention is therefore the person's scale and complete geometry, not removal of the scene.

6.3 Representation dominates the conventional ConvNet control

Under matched person-centric preprocessing and target training, frozen DINOv2-B reaches 0.8476 validation macro-F1 and ConvNeXt-S reaches 0.7726. SigLIP2-B reaches 0.8449 and is statistically tied with DINOv2-B in the paired comparison. DINOv2-B is selected because it combines the best point result with a simpler vision-only feature path and strong multiview complementarity.

6.4 Structure and geometry help in combination

The pose oracle reaches only 0.6954, while the visual-plus-geometry DINO models exceed 0.84. Geometry alone does not solve the task. It becomes useful as a conditioning signal for reconciling tight and contextual visual evidence. Separately, the matched factorized comparison shows that encoding seated/upright and stationary/locomoting structure improves the decision rule even when every input feature is fixed.

6.5 Partial fine-tuning has limited marginal value here

LP-FT with person-safe AugMix reaches 0.8580, within the uncertainty interval of the 0.8594 frozen stack. This supports a practical conclusion for this data regime: high-quality person-centric DINO features plus a controlled classifier capture nearly all of the measured benefit, while top-block tuning increases training cost without a reliable accuracy margin.

7. Engineering controls

- DINOv2 and SigLIP2 model revisions are pinned rather than resolved from a moving default branch.
- Cross-validation, bootstrapping, and few-shot sampling use fixed seeds.
- People from one source image remain together in cross-validation and uncertainty resampling.
- Every experiment checks the protocol hash and records zero official-test rows read during development.
- The final stack refit reproduces the selected validation probabilities with maximum absolute difference 0.0 before using all development rows.
- The label-free test feature artifacts are hash-checked before evaluation.
- The official-test evaluator is idempotent and rejects a changed cached manifest, selection lock, feature artifact, checkpoint, or implementation source.
- Portable evidence is generated by one exporter and inventoried with SHA-256 hashes.

The repository's lint, unit-test, and evidence validators pass against the exported package. The test gate and portable evidence are checked again during release validation.

8. Scope and remaining questions

The results establish a reproducible person-level endpoint under this action-tag mapping. Three extensions would add the most scientific value:

- 1 Independently relabel a balanced sample of sitting, standing, walking, and running from pixels, with adjudication and inter-rater agreement. This would separate model error from source-tag semantics.
- 2 Add a temporal dataset. Still images cannot resolve motion when pose and context are visually compatible with both standing and locomotion.
- 3 Repeat the locked protocol on another person-centric dataset with different image sources. That would test whether the scale-conditioned gain transfers beyond COCO imagery.

The controlled representation screen was limited to the revision-pinned DINOv2-B, ConvNeXt-S, and SigLIP2-B checkpoints declared in the protocol.

9. Reproduction map

The compact evidence lives under `results/vcoco_v2/`. The core sequence is:

- 1 `tools/build_vcoco_v2_manifests.py` builds the factorized split manifests.
- 2 `tools/lock_vcoco_v2_protocol.py` fixes the endpoint and test policy.

- 3 `experiments/cache_vcoco_v2_features.py` extracts revision-pinned development features.
- 4 The screen, few-shot, factorized, neural, background, and mechanism scripts produce development evidence.
- 5 `experiments/fit_vcoco_v2_final_stack.py` verifies selection replay and refits all development rows.
- 6 `tools/lock_vcoco_v2_final_selection.py` binds the champion and evaluation sources.
- 7 `experiments/cache_vcoco_v2_final_test_features.py` performs label-free test feature extraction.
- 8 `experiments/evaluate_vcoco_v2_final_test.py` opens the test labels once and writes the final evidence.
- 9 `tools/export_vcoco_v2_results.py` creates the portable package.

Run `python -m ruff check .`, `python -m pytest`, and `python tools/validate_repository.py` before committing a release.

10. Conclusion

The original external gap was largely recoverable without a new backbone. The largest improvement came from treating the person instance, its scale, and its local context as first-class inputs. A structured classifier added a smaller but independently supported gain. Partial fine-tuning and stronger augmentation remained competitive but did not beat the frozen stack by the locked margin. On the official test set, the final system improved macro-F1 by 15.9 points, removed most of the historical small-person penalty, and produced calibrated confidence that supports useful abstention.

References

- 1 S. Gupta and J. Malik. [Visual Semantic Role Labeling](#), 2015.
- 2 M. Oquab et al. [DINOv2: Learning Robust Visual Features without Supervision](#), 2023.
- 3 Z. Liu et al. [A ConvNet for the 2020s](#), CVPR 2022.
- 4 M. Tschannen et al. [SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features](#), 2025.
- 5 A. Kumar et al. [Fine-Tuning can Distort Pretrained Features and Underperform Out-of-Distribution](#), ICLR 2022.
- 6 D. Hendrycks et al. [AugMix: A Simple Data Processing Method to Improve Robustness and Uncertainty](#), 2020.